

YUJIAN LIU

✉ yujianliu@ucsb.edu · 🔗 <https://yujianll.github.io>

RESEARCH INTERESTS

Large language model reasoning, agentic applications, trustworthiness, diffusion models

EDUCATION

University of California, Santa Barbara, Santa Barbara, CA Sep 2022 - Now
Ph.D. student in Computer Science, advised by **Prof. Shiyu Chang**

University of Michigan, Ann Arbor, Ann Arbor, MI Sep 2018 - May 2022
M.S. in Computer Science, advised by **Prof. Lu Wang** and **Prof. Rada Mihalcea**
B.S. in Computer Science, Minor in Mathematics
GPA: 4.00/4.00

Shanghai Jiao Tong University, Shanghai, China Sep 2016 - Aug 2018
B.S. in Electrical and Computer Engineering
GPA: 3.75/4.00

WORK EXPERIENCE

AMD GenAI, Santa Clara, CA Jun 2025 - Sep 2025
Research Intern
Primary mentor: Dr. Ze Wang

MIT-IBM Watson AI Lab, Cambridge, MA Jun 2024 - Sep 2024
Research Intern
Primary mentor: Dr. Yang Zhang

Adobe Research, San Jose, CA Jun 2023 - Sep 2023
Research Intern
Primary mentor: Dr. Tong Yu

PUBLICATIONS

* co-first author

Conference Publications

- [1] **Yujian Liu**, Shiyu Chang, Tommi Jaakkola, and Yang Zhang. “Fictitious Synthetic Data Can Improve LLM Factuality via Prerequisite Learning”. **ICLR, 2025**. [paper] [code].
- [2] **Yujian Liu**, Jiabao Ji, Tong Yu, Ryan Rossi, Sungchul Kim, Handong Zhao, Ritwik Sinha, Yang Zhang, and Shiyu Chang. “Augment before You Try: Knowledge-Enhanced Table Question Answering via Table Expansion”. **EMNLP-Findings, 2025**. [paper] [code].
- [3] Li An, **Yujian Liu**, Yepeng Liu, Yang Zhang, Yuheng Bu, and Shiyu Chang. “Defending LLM Watermarking Against Spoofing Attacks with Contrastive Representation Learning”. **COLM, 2025**. [paper] [code].
- [4] **Yujian Liu**, Yang Zhang, Tommi Jaakkola, and Shiyu Chang. “Revisiting Who’s Harry Potter: Towards Targeted Unlearning from a Causal Intervention Perspective”. **EMNLP, 2024, Oral**. [paper] [code].
- [5] Jiabao Ji, **Yujian Liu**, Yang Zhang, Gaowen Liu, Ramana Rao Kompella, Sijia Liu, and Shiyu Chang. “Reversing the Forget-Retain Objectives: An Efficient LLM Unlearning Framework from Logit Difference”. **NeurIPS, 2024**. [paper] [code].
- [6] Bairu Hou, **Yujian Liu**, Kaizhi Qian, Jacob Andreas, Shiyu Chang, and Yang Zhang. “Decomposing Uncertainty for Large Language Models through Input Clarification Ensembling”. **ICML, 2024, Oral** [paper] [code].

- [7] **Yujian Liu**, Yang Zhang, Tommi Jaakkola, and Shiyu Chang. “Correcting Diffusion Generation through Resampling”. **CVPR, 2024, Highlight**. [paper] [code].
- [8] Qiucheng*, **Yujian Liu***, Handong Zhao, Trung Bui, Zhe Lin, Yang Zhang, and Shiyu Chang. “Harnessing the Spatial-Temporal Attention of Diffusion Models for High-Fidelity Text-to-Image Synthesis”. **ICCV, 2023**. [paper] [code].
- [9] Qiucheng Wu, **Yujian Liu**, Handong Zhao, Ajinkya Kale, Trung Bui, Tong Yu, Zhe Lin, Yang Zhang, and Shiyu Chang. “Uncovering the Disentanglement Capability in Text-to-Image Diffusion Models”. **CVPR, 2023**. [paper] [code].
- [10] **Yujian Liu**, Xinliang Frederick Zhang, Kaijian Zou, Ruihong Huang, Nicholas Beauchamp, and Lu Wang. “All Things Considered: Detecting Partisan Events from News Media with Cross-Article Comparison”. **EMNLP, 2023**. [paper].
- [11] **Yujian Liu***, Laura Biester*, and Rada Mihalcea. “Improving Mental Health Classifier Generalization with Pre-Diagnosis Data”. **ICWSM, 2023**. [paper]
- [12] **Yujian Liu***, Xinliang Frederick Zhang*, David Wegsman, Nicholas Beauchamp, and Lu Wang. “POLITICS: Pretraining with Same-story Article Comparison for Ideology Prediction and Stance Detection”. **NAACL-Findings, 2022**. [paper] [code].

Preprints

- [1] **Yujian Liu**, Jiabao Ji, Yang Zhang, Wenbo Guo, Tommi Jaakkola, Shiyu Chang. “HarnessLLM: Automatic Testing Harness Generation via Reinforcement Learning”. **DL4C@NeurIPS, 2025**. [paper] [code].
- [2] **Yujian Liu**, Ze Wang, Hao Chen, Ximeng Sun, Xiaodong Yu, Jialian Wu, Jiang Liu, Emad Barsoum, Zicheng Liu, Shiyu Chang. “Learning from Online Videos at Inference Time for Computer-Use Agents”. **arXiv, 2025**. [paper] [code].
- [3] Li An, **Yujian Liu**, Yepeng Liu, Yuheng Bu, Yang Zhang, and Shiyu Chang. “A Reinforcement Learning Framework for Robust and Secure LLM Watermarking”. **arXiv, 2025**. [paper] [code].
- [4] Bairu Hou, Yang Zhang, Jiabao Ji, **Yujian Liu**, Kaizhi Qian, Jacob Andreas, and Shiyu Chang. “ThinkPrune: Pruning Long Chain-of-Thought of LLMs via Reinforcement Learning”. **arXiv, 2025**. [paper] [code].
- [5] Chung Hoon Hong, Yuan Liang, Sagnik Sinha Roy, Arushi Jain, Vihang Agarwal, Ryan Draves, Zhizhuo Zhou, William Chen, **Yujian Liu**, Martha Miracky, Lily Ge, Nikola Banovic, and David Jurgens. “Audrey: A Personalized Open-Domain Conversational Bot”. **arXiv, 2020**. [paper].

TEACHING AND MENTORSHIP

Teaching

- **Teaching Assistant** at University of California, Santa Barbara Fall 2025, 2024, 2023, 2022
Introduction to Deep Learning, Machine Learning, Introduction to Optimization
- **Teaching Assistant** at University of Michigan, Ann Arbor Winter 2022, 2021
Introduction to Natural Language Processing

Mentorship

- Early Research Scholars Program (University of California, Santa Barbara) Fall 2023-Spring 2024
Undergraduate students: Lindsey Wen, Riya Gupta, Peiyang Song, Ethan Solomon
Project: Uncertainty in large language models
- Early Research Scholars Program (University of California, Santa Barbara) Fall 2022-Spring 2023
Undergraduate students: Liliana Nguyen, Hugo Lin, Shamita Gurusu, Edwin Yee, Wesley Truong
Project: Automatic prompt generation for diffusion models

SELECTED AWARDS

James B. Angell Scholar, University of Michigan: March 2020
 University Honors, University of Michigan: Fall 2018 - Winter 2020
 University of Michigan College of Engineering Dean’s Honor List: Fall 2018 - Winter 2020
 SJTU Undergraduate Scholarship: Dec 2017